

CredenceLedger: A Permissioned Blockchain for Verifiable Academic Credentials

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Abstract—Blockchain, the underlying technology that powers cryptocurrencies such as Bitcoin and Ethereum, is gaining so much attention from different industry stakeholders, governments and research communities. Its application is extending beyond cryptocurrencies and has been exploited in different domains such as finance, E-commerce, Internet of Things (IoT), healthcare, and governance. Some key attributes of the technology are decentralization, immutability, security and transparency. This paper aims to describe how permissioned Blockchain can be applied to a specific educational use case — decentralized verification of academic credentials. The proposed Blockchain-based solution, named ‘CredenceLedger’, is a system that stores compact data proofs of digital academic credentials in Blockchain ledger that are easily verifiable for education stakeholders and interested third party organizations.

Index Terms—Blockchain, Cryptocurrency, Decentralized Verification, Stream, Mining

I. INTRODUCTION

As early as birth, there is a need to document people and ensure that these documents are verifiable. These documents that prove authenticity of what has been achieved continue to pile up as people move towards gaining educational attainments, training certifications and work experience records. Use of fake documents has been a problem for so many years. Nowadays, forging ‘paper-based’ documents to produce replicas of the authentic or official copies can be done in just a couple of hours. At worst, in the absence of a security mechanism in place, fraudulent copies of ‘digital-based’ credentials can be fabricated at enormous speed.

In 2009, Judge Jenny Lind Aldecoa-Delorino filed a case against a court clerk who falsified the school records to qualify for the job [1]. At MIT, a similar matter forced former Dean of Admissions, Marilee Jones, to quit her job when she admitted that for nearly three decades at the institution, she represented herself as having three degrees from different New York institutions when in fact, she had not achieved any of these [11]. Such shortcomings could have been avoided if, in the case of Marilee Jones, MIT officials knew her academic credentials were verifiable.

While it is easy to counterfeit credentials almost instantaneously, verification of authenticity is more challenging. This also shows that the relationship between copying and verifying are inversely proportional processes.

This is a problem that, ideally, has to be addressed by education institutions and interested third parties.

In the Philippines, finding a job is a fierce competition. Normally, a prospective employer will look for the candidate’s academic credentials before making a hiring decision. Most companies searching for skilled or professional workforce, require some proof of qualification. This proof is very often paper-based certified copy of the candidate’s credentials which can involve an additional task of proving or disproving the copy’s authenticity. As a further matter, nothing is stopping the candidates from forging their education documents, which is easier than finding a job [3]. A survey conducted by Career Builder shows that 33 percent of the respondents lie about their academic degree [2] which is relatively amass proportion of the 53 percent fraudulent resumes employer receives worldwide [4].

The ‘Recto University’ [3], Philippines’ very own version of Diploma Mill [5], offers fake documents for a fee [6] [7] despite committing a criminal offence [8]. These ‘Recto U’ credentials are not verifiable but have already reached the international communities [9] [10]. A ‘Recto U’ graduate would have just to wait a few hours to get hold of an academic degree, and maybe few days or weeks of getting a job. Because it is way harder to verify the person’s credentials, the employee could work for many years or even be promoted to top position such as a decision-maker, a leader, or an official of the government or academic institution.

Any education ecosystem is composed of multiple stakeholders, both from public and private sectors, such as regulatory boards, government, admission committees, testing companies, accreditors, educators and students who constitute the institutions. Each institution may have multiple aggregators transacting academic documents. In many instances, these documents are shared by the recipients to another institution or some third parties such as an employer, for legal and beneficial purposes. The third party or institution needs to verify the authenticity of a document by contacting the original issuer and asking the latter whether they really issued the certificate. Moreover, the original issuer might consult their centralized or local databases of certificate records. Otherwise, if the original issuer does not have a database, the traditional way of checking the security features built into the paper-based certificate will

be applied. Therefore, document verification is a slow process and can be hard or easy depending on the institution. In the same manner, a media company [13] describes the hardships and laborious effort they invest in fact-checking the academic degrees of some politicians.

The solution proposed in this paper for the decentralized verification of academic credentials relies on a permissioned Blockchain-based technology specifically the open source Multichain [17]. Blockchain is a fast-growing technology that powers Bitcoin and other cryptocurrencies, forecasted to disrupt any traditional services because of the decentralized and distributed nature. Blockchain can be used as a transparent and trusted system for securing, sharing, and verifying academic credentials. Blockchain also represents an opportunity for third parties such as employers, to independently and privately verify shared academic proofs.

II. BLOCKCHAIN TECHNOLOGY

Blockchain is a decentralized ledger that records history of information from a group of validated transactions called 'block'. Each block is timestamped and cryptographically linked to a previous block forming a chain of immutable records that are distributed, shared and maintained across multiple nodes on a P2P network. In order to add a block into an ongoing chain, an extremely difficult mathematical proof called 'mining' is required to confirm transactions. A block can only be verified based on synchronous consensus of the miners.

A. Blockchain 1.0: Bitcoin Cryptocurrency and Public Blockchain

Bitcoin is the first, largest and, arguably, the most successful implementation of Blockchain. It is also the first cryptocurrency continuously operating on public Blockchain invented in 2008 [12], by Satoshi Nakamoto, a pseudonym.

1) *Blockchain Data Structure*: Blockchain is a decentralized append-only database or 'ledger' for cryptographically linked blocks of transactions, each pointing to a previous or parent block. A block that has no parent is the 'genesis block' which also constitutes the first and foundation block in the Blockchain. Likewise, the most recently added block in an ongoing chain is called 'top block'. Each block is made up of a header and body. A 'block header' consists of three metadata: (1) previous block hash; (2) mining metadata — difficulty, timestamp and nonce; and (3) merkle root. Block aggregates and validates transactions for inclusion in the 'block body'. A block's primary identifier is a 32-byte cryptographic hash called 'header hash'. Its secondary identifier is the 'height' or the distance between the 'genesis block' and 'top block'. The 'merkle root' is the summary of all the transactions in a block. It derives from a 'merkle tree', also known as 'binary hash tree', a branching data structure used to efficiently summarize and verify the integrity of large sets of data [15].

2) *Blockchain Network and Protocols*: Blockchain is a purely decentralized P2P consensus network where all computers or nodes that participate are peers to each other. Nodes in a Blockchain network may take one or more different roles from the four main protocols they can support: database, mining, routing, and wallet services. Full nodes maintain a complete up-to-date copy of the Blockchain database. Wallets verify transactions and can be part of a full node or a node running on constrained devices such as mobile phones which maintain a subset of Blockchain data and can verify transactions using a method called Simple Payment Verification (SPV). Mining nodes create new blocks and protect the network by participating in the synchronous consensus. Routing nodes run other protocols such as mining pool protocol [15], for other nodes.

3) *Bitcoin's Transactional Model*: Transaction is the most important component of cryptocurrency. It is based on Bitcoin's transactional model invented by Satoshi Nakamoto. Every Bitcoin transaction has a set of inputs and a set of outputs. Full nodes track all available and spendable outputs in the Blockchain. These transaction outputs called Unspent Transaction Output (UTXO), create spendable sum of Bitcoin or available 'balance' for the owner to spend in the future. If a wallet or SPV node creates a new (spend) transaction, a UTXO set from its spendable balance will become the transaction input that will provide proof of ownership via an unlocking script. The newly created transaction will be processed by some mining nodes in their candidate block. Only one of the candidate blocks can be added to the ongoing Bitcoin Blockchain after which, its miner receives the 'block reward' and transaction fees. In Bitcoin, transaction fees are calculated based on the transaction size wherein a byte costs 1 Satoshi or 0.00000001 BTC [12], and mining reward is currently 12.5 BTC [33] because of the 4-year halving schedule.

4) *Mining and Consensus Algorithms*: Mining is the decentralized security mechanism of public Blockchain by which transactions are cleared and verified. Mining node or 'miner' validates block of transactions and records them on the decentralized ledger. A new block is mined every 10 minutes on average for Bitcoin and every 17 seconds on average for Ethereum.

Decentralized consensus is the most important function of mining. It is the global agreement from thousands of independent asynchronous nodes achieved without a central authority.

Bitcoin implements Proof-of-Work (PoW) by incrementing a nonce in the block until a value is found to give the block's hash the required zero bits [12]. Similarly, Ethereum's Proof-of-Stake (PoS) is another category of consensus algorithm for public Blockchains that depends on a miner's economic stake or available cryptocurrency in the network.

Mining as an incentive system, provides two types of reward in return for the security provided by the miners: transaction fees and newly minted coins. To earn this reward, miners compete to solve an extremely difficult pointless mathematical

problem based on a cryptographic hash algorithm, that consumes a lot of electricity. The estimated difficulty level, at the time of this writing, is 25,517,998.11 TH/s (tera hashes or trillions of hashes per second) [33], which is regularly adjusted by the public Blockchain network to maintain a steady rate of minting the cryptocurrency.

5) *Cryptography and Security*: Asymmetric cryptography is used when storing transaction data in a Blockchain by digitally signing them so that only intended users can open or read the contents, thus, forming a chain of digital signatures. This process makes it possible for anyone to verify every signature on every transaction while ensuring that only the owners of private keys can produce valid signatures.

B. Blockchain 2.0: Smart Contracts and New Cryptocurrencies

There are many next-generation public Blockchains developed and launched [26]. A 'smart contract' is a general-purpose program that can be executed on a Blockchain. It was popularized by Ethereum [27], a public Blockchain associated to a cryptocurrency called 'ether'.

C. Blockchain 3.0: Beyond Cryptocurrencies

While Bitcoin and Ethereum top the cryptocurrency market [26], many private or permissioned Blockchains such as Multichain [17], provided more Blockchain potential use cases that impacted many domains such as financial institutions, governments, corporates and cross-industries [14].

D. Blockchain 4.0: Blockchain-as-a-Service (BaaS)

Open Source communities, technology giants and start-up companies are investing heavily in Blockchain solutions. IBM developed its own BaaS based on Hyperledger [25], an open source Blockchain framework that resulted from global collaborative efforts initiated by the Linux Foundation and 130+ technology leaders. Likewise, Microsoft expanded their support to Ethereum-based BaaS powered by Azure.

E. Existing Blockchain-based Solutions for Academic Credentials

Grech et. al.¹ reported several use cases where Blockchain technology can be adopted in education context [18]. Decentralized verification of academic credentials is just one of the many potential use cases. Relatively, listed below are some solution vendors already offering Blockchain-based services to education sector that may be applicable to decentralized verification of academic credentials:

1) *BlockCert*: BlockCert is an open source software that any university or institution can use to issue credentials using Bitcoin Blockchain. It was developed by MIT Media Lab in collaboration with Learning Machine. With the addition of the wallet app, students' information is encrypted. They can prove ownership of their diploma through the generation of

a unique numerical identifier so they can quick share their virtual certificates with potential employers without involving an intermediary. Third parties can verify the legitimacy of the diploma by pasting the Uniform Resource Locator (URL) of the certificate into an MIT-hosted portal [16].

2) *Sony Global Education*: The Sony Global Education uses IBM's Hyperledger to support different education institutions to record educational data and issue digital transcripts. The solution provides an Application Programming Interface (API) to customize the attributes data collections of students' records and control usage rights for educational institutions [29].

3) *Open Certificate*: Open Certificate is a certificate issuance platform by Attores, wherein a university or institution can issue degrees or diplomas directly on the Blockchain in the form of a smart contracts built on Ethereum's computing platform. The platform also uses Inter Planetary File System (IPFS), a Dropbox-like decentralized storage system [28].

4) *Gradbase*: Gradbase is a qualification records verification system that utilizes Bitcoin's Blockchain [30]. The system provides a modifiable transaction model that can verify the qualification of a person by providing data, without requiring its users to login or register an account.

5) *Stampery*: Stampery uses Bitcoin Blockchain to timestamp and verify data. It also uses Ethereum and Bitcoin proof using routing keys and the previously created bindings. According to Stampery, the method provides Blockchain anchoring that is scalable and cost effective [31].

III. PRELIMINARIES

A. Paper-based Academic Records

Paper documents have continued popularity because of the credibility that they represent [32]. Businesses also inspire the use of paper especially on documents that can be used as evidence or proof. Unfortunately, these can be forged in places like "Recto U".

B. Digital Academic Records

Digital preservation of documents and records are being practiced, but there is a low confidence on its quality mainly because of system errors, crashes and hacking [32]. Digital data also requires management information systems that are secure, efficient and scalable for the changing needs of the academic institutions.

C. Centralized Databases

A database is a structured repository of information. This can be a traditional relational database, wherein data are organized into tables. Or it can be the trendier NoSQL variety, which works more like a key-value file system or dictionary. Databases are modified via transactions which represent a set of changes that must be accepted or rejected as a whole.

In the education context, there needs to be more than one entity involved in generating the transactions that modify the academic records on the database. If multiple entities are

¹Alexander Grech, Anthony Camilleri and Andreia Inamorato released in 2017 a study for the Joint Research Center of the European Commission entitled 'Blockchain in Education'.

writing to the database, there also needs to have some degree of mistrust between those writers. For example, one user is not willing to let another modify database entries which it owns. A well-known solution to this problem of enabling a database with multiple non-trusting writers is a trusted intermediary or simply, a centralized database.

Many systems are filled with databases of this nature, such as the ledger of accounts in a bank. The bank controls the database and ensures that every transaction is valid and authorized by the customer.

D. Permissioned Blockchain

Permissioned Blockchain inherited many good features from public or permissionless Blockchain with added functionalities ideal to closed or regulated systems that operate according to a set of rules or permissions. It also addresses several shortcomings of a public Blockchain related to: consensus mechanism and mining risks; access and participation; and privacy.

The participating Blockchain nodes are added on a 'by invitation' process which is a common interest in the network. Correspondingly, the Blockchain ledger will only contain transaction data relevant to those participants. Arguably, permissioned Blockchain tends to be more efficient in terms of: high transaction throughputs, and low costs and resources consumption.

E. CredenceLedger

CredenceLedger is a proposed permissioned Blockchain-based platform for decentralized verification of academic credentials.

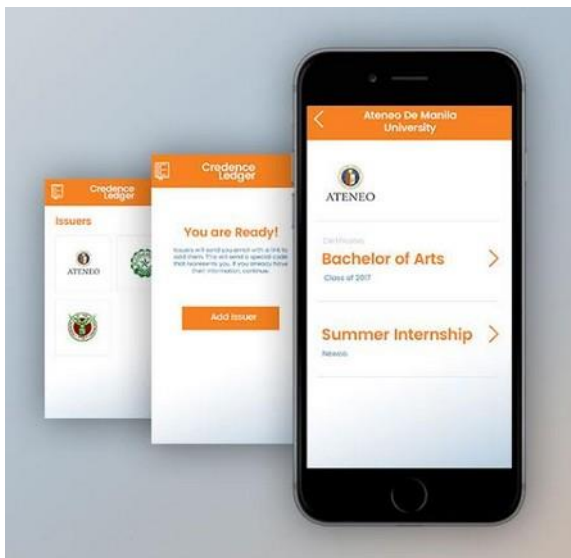


Fig. 1. CredenceLedger Mobile Application.

This software platform is based on Multichain with the addition of the CredenceLedger mobile application, offering the students the option to be awarded authentic digital

version of their credentials when they graduate, in addition to receiving the paper-based version of the credentials. By using the mobile app, the students can quickly access a digital credential that can be easily verified by interested third parties such as employers, to ensure its authenticity and shared on social media. The digital credentials are protected using Blockchain's built-in security features which means that limited information (compact data proofs) will be available to interested third parties. Other information is encrypted and will only be available upon request subject to recipient's approval. Decentralized verification of authenticity and recipient's full control over their privacy are the two important aspects of this solution.

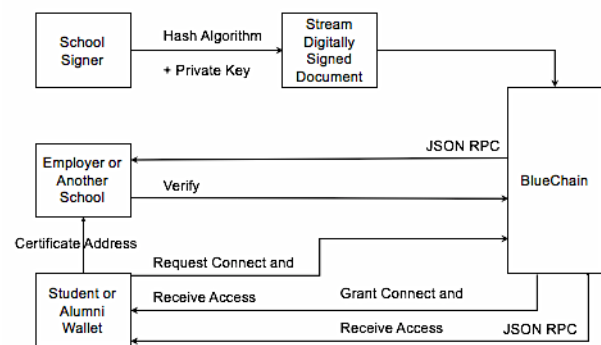


Fig. 2. CredenceLedger Architecture.

All privileges provided using the software can be modified using the transaction metadata depending on what permissions available to its users. These permissions can be divided into three categories: 'Low Risk' such as connect, send and receive permissions; 'Medium Risk' such as issue, create, activate permissions; and 'High Risk' such as mine and admin permissions.

CredenceLedger makes use of streams² so that there would be no need to transact using cryptocurrency. Streams are represented by a Blockchain transaction and integrated with the permissions put in place in the network. Stream is an abstraction of a database for Blockchain.

Mining is implemented using a scheme called 'mining diversity', which is constrained by $0 \leq \text{mining diversity} \leq 1$ [17]. This avoids monopoly in the mining process by implementing round-robin validation scheme. According to Greenspan, mining diversity is suggested to be set at 0.75 to keep both technical failures and malicious collusion below 0.001 percent based on the following equation:

²Hexadecimal value that uses key value pair

$$\begin{aligned}
Pr(F) &= Pr(s-1; m, f) \\
&= \binom{m}{s-1} f^{s-1} (1-f)^{m-s-1} \\
\text{for} \quad \binom{m}{s-1} &= \frac{m!}{(s-1)!(m-s-1)!}
\end{aligned} \tag{1}$$

IV. ANALYSIS

A. Paper-based vs. Digital Academic Records

The paper format issued by schools and universities has a minimum turnaround time of three days [20] up to one month [21]. If the student or former alumni would like to study or work abroad, these documents should be authenticated through Department of Foreign Affairs, a process called Red Ribbon which usually takes four days [23]. If the purpose of the documents is for local use, the processing time is 10 minutes [22]. Certificates and documents from schools are in paper format even though digitalization of the school system is being practiced. The characteristics of the paper format that could be improved in the proposed CredenceLedger are shown in Table I.

TABLE I
CHARACTERISTICS OF PAPER-BASED AND DIGITAL CREDENTIALS

Characteristics of a Certificate	Paper-based	Digital
Can be forged	Yes	No
Manual Verification	Yes	No
Can be revoked	No	Yes
Can be destroyed	Yes	No

B. Centralized Databases vs. Blockchain

A centralized server has all the read-write access that could replace all the data that the database holds while a centralized client only has the ability to change and read its own data. Also, a centralized client and server is vulnerable to hacking which if left unnoticed will become a permanent record [24]. In Blockchain, anyone could read everyone's data and write on its own but they cannot change or replace data of everyone else as shown in Table II.

TABLE II
CHARACTERISTICS OF BLOCKCHAIN AND CENTRALIZED DATABASES [19]

Access	Blockchain	Centralized Client	Centralized Server
Read mine	Yes	Yes	Yes
Read all	Yes	No	Yes
Write mine	Yes	Yes	Yes
Write all	No	No	Yes

C. Permissionless vs. Permissioned Blockchain

The main difference of a permissioned Blockchain, like the proposed CredenceLedger, from public or permissionless Blockchain are the following features shown in Table III. In CredenceLedger, permissions are added to every transaction's metadata using stream.

TABLE III
FEATURES OF PERMISSIONLESS AND PERMISSIONED BLOCKCHAIN [18]

Feature	Permissioned	Permissionless
Open to everyone	No	Yes
Scalability	Yes	No
Low storage cost	Yes	No
Smaller chain	Yes	No

D. CredenceLedger vs. Other Blockchain-based Solutions

It could be observed on Table IV that the available Blockchain implementation for education is using Bitcoin's transaction for asset. On the other hand, the proposed CredenceLedger implements streaming through Multichain for the privacy of the school documents. All other source code is proprietary except for BlockCert and CredenceLedger.

TABLE IV
EXISTING SOFTWARE IMPLEMENTATION OF BLOCKCHAIN IN EDUCATION

Software Solution	Blockchain Implementation	Require Cryptocurrency	Streaming Data	Open Source
BlockCert	Blockchain through the mining process as Bitcoin transaction.	Yes	No	Yes
Open Certificate	Blockchain in form of a Smart Contract.	Yes	No	No
Gradbase	Bitcoin Blockchain with an option for custom transaction.	Yes	No	No
Stampery	Bitcoin Blockchain and Ethereum proof	Yes	No	No
Credence Ledger	Multichain with multisignature.	No	Yes	Yes

V. CONCLUSIONS

CredenceLedger, a permissioned Blockchain-based platform for centralized verification of academic credentials is proposed and analyzed. The major observations are as follows:

1. The proposed system can award digital version of student's credentials that are easily verifiable and without the need to transact via public Blockchain using cryptocurrency. It also presents an opportunity for third parties such as employers, to independently and privately verify shared

academic proofs.

2. That permissioned Blockchain can be an effective distributed system for shared database or 'ledger' in which multiple entities verify data without the need for a centralized system.

3. That permissioned Blockchain is an efficient system for managing digital academic credentials mainly in terms of: high transaction throughputs, and low costs and resources consumption.

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